Robert Riley Nerem

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RESEARCH

My research in neural algorithmic reasoning seeks to understand and develop algorithmic capabilities in machine learning. I study when neural networks learn algorithms and generalize across graph sizes and distributions. I combine theoretical analysis with algorithmic alignment to design learning methods for combinatorial optimization, motivated by applications in biology and chip design.

EDUCATION

Doctor of Philosophy, Data Science (in progress) — September 2022–Present
University of California San Diego, San Diego, CA
Advisor: Prof. Yusu Wang, Halıcıoğlu Data Science Institute

Master of Science, Physics — May 2022
University of Calgary, Calgary, AB
Supervisor: Prof. Barry Sanders, Department of Physics and Astronomy
Co-supervisor: Prof. Peter Høyer, Department of Computer Science

Bachelor of Science, Physics — December 2018
Montana State University, Bozeman, MT
Honors Degree, *summa cum laude*
Minors: Computer Science, Mathematics

PUBLICATIONS

* Indicates a primary author in addition to the first-listed author.

Conference Papers

S. Wittig*, A. Vasileiou*, **R.R. Nerem***, T. Stoll, F. Geerts, Y. Wang, and C. Morris. Which Algorithms Can Graph Neural Networks Learn? International Conference on Machine Learning (ICML), PMLR **306**, 135232–135307, 2026. **Spotlight; Oral**.

R.R. Nerem, S. Chen*, S. Dasgupta, and Y. Wang. Graph Neural Networks Extrapolate Out-of-Distribution for Shortest Paths. Conference on Learning Theory (COLT), PMLR **336**, 5273–5331, 2026.

R.R. Nerem, Z. Luo, A. Rafiey, and Y. Wang. Differentiable Extensions with Rounding Guarantees for Combinatorial Optimization over Permutations. Advances in Neural Information Processing Systems (NeurIPS), **38**, 2025.

A.B. Kahng, **R.R. Nerem**, Y. Wang, and C.-Y. Yang. NN-Steiner: A Mixed Neural-Algorithmic Approach for the Rectilinear Steiner Minimum Tree Problem. Proceedings of AAAI, **38**(12), 13022–13030, 2024.

D.V. Reddy, **R.R. Nerem**, A.E. Lita, S.W. Nam, R.P. Mirin, and V.B. Verma. Exceeding 95% System Efficiency Within the Telecom C-Band in Superconducting Nanowire Single Photon Detectors. CLEO: QELS—Fundamental Science, FF1A.3, 2019.

Preprints

R.R. Nerem, M. Black, A. Vasileiou, C. Morris, and Y. Wang. The Expressive Power of Random-Feature MPNNs Across Graph Sizes. arXiv:2610.12809, 2026.

R.R. Nerem, P. Singh, C. Ward, and Y. Wang. Algorithmically Aligned Neural Agglomerative Tree Construction. arXiv:2610.07271, 2026.

R.R. Nerem, P. Crawford-Kahrl, B. Cummins, and T. Gedeon. A Poset Metric from the Directed Maximum Common Edge Subgraph. arXiv:1910.14638, 2019 (revised 2020).

Workshop Contributions

Y. He, **R.R. Nerem**, T. Stoll, S. Cantürk, D. Georgiev, C. Qian, S. Wittig, F. Geerts, S. Jegelka, E. Vitercik, Y. Wang, N. Karalias, and C. Morris. Neural Algorithmic Reasoning Must Explain When Neuralization Adds Value. 3rd AI for Math Workshop, ICML, 2026.

Journal Articles

R.R. Nerem and D.R. Gaur. Conditions for Advantageous Quantum Bitcoin Mining. Blockchain: Research and Applications, **4**(3), 100141, 2023.

A. Alase, **R.R. Nerem**, M. Bagherimehrab, P. Høyer, and B.C. Sanders. Tight Bound for Estimating Expectation Values from a System of Linear Equations. Physical Review Research, **4**(2), 023237, 2022.

P. Crawford-Kahrl*, **R.R. Nerem***, B. Cummins, and T. Gedeon. Genetic Networks Encode Secrets of Their Past. Journal of Theoretical Biology, **541**, 111092, 2022.

D.V. Reddy, **R.R. Nerem**, S.W. Nam, R.P. Mirin, and V.B. Verma. Superconducting Nanowire Single-Photon Detectors with 98% System Detection Efficiency at 1550 nm. Optica, **7**(12), 1649–1653, 2020.

L.M. Smith, F.C. Motta, G. Chopra, J.K. Moch, **R.R. Nerem**, B. Cummins, K.E. Roche, C.M. Kelliher, A.R. Leman, J. Harer, T. Gedeon, N.C. Waters, and S.B. Haase. An Intrinsic Oscillator Drives the Blood Stage Cycle of the Malaria Parasite Plasmodium falciparum. Science, **368**(6492), 754–759, 2020.

R.R. Nerem and D. James. Charged Particle Pickup Tube Detector Used in Experimental Applications and Classroom Demonstrations. The Physics Teacher, **58**(3), 200–203, 2020.

E. Berry, B. Cummins, **R.R. Nerem**, L.M. Smith, S.B. Haase, and T. Gedeon. Using Extremal Events to Characterize Noisy Time Series. Journal of Mathematical Biology, **80**, 1523–1557, 2020.

RESEARCH EXPERIENCE

Computation

PhD Research Fall 2022–Present
University of California San Diego, San Diego, CA
Prof. Yusu Wang, Halıcıoğlu Data Science Institute

Established GNN expressivity and size-generalization results; developed differentiable permutation objectives and algorithmically aligned models.

Foundational AI Research Intern (part time) August–December 2025
Intel, AI Lab
Manager: Somdeb Majumdar; collaborator: Shubha Kharel

Designed machine learning models for combinatorial optimization aligned with simulated annealing and inspired by diffusion approaches.

Quantum Researcher Fall 2021–Fall 2022
Quantum Algorithms Institute, Surrey, BC

Investigated quantum machine learning algorithms for medical biology.

Master's Research Fall 2019–May 2022
University of Calgary, Calgary, AB
Prof. Barry Sanders & Prof. Peter Høyer, Institute for Quantum Science and Technology

Constructed quantum expectation-value algorithms and proved lower bounds; assessed quantum methods for Bitcoin mining and pathogen identification.

Research Assistant Fall 2018–Summer 2019
Montana State University, Bozeman, MT
Prof. Tomas Gedeon & Dr. Breschine Cummins, Mathematics

Formulated graph-based metrics for gene-expression time series and devised algorithms to identify conserved structure in evolving genetic networks.

Experiment

Research Assistant Summer 2018
National Institute of Standards and Technology, Boulder, CO
Dr. Sae Woo Nam, Faint Photonics Group

Automated precision measurements of single-photon detector efficiency and integrated photonic devices using Python.

Research Assistant Fall 2016–Spring 2018
Montana State University, Bozeman, MT
Prof. Rufus Cone & Dr. Charles Thiel, Physics

Built scanning Fabry–Perot interferometers and characterized crystals spectroscopically for quantum information applications.

Research Assistant Summer 2015, Summer 2016
University of Colorado, Boulder, CO
Prof. Zoltan Sternovsky & Dr. David James, Institute for Modeling Plasma, Atmospheres and Cosmic Dust

Engineered a pickup detector and integrated photomultiplier instrumentation to measure particle charge, velocity, and acceleration.

TEACHING EXPERIENCE

Teaching Assistant Fall 2022
University of California San Diego, San Diego, CA
MATH 154, Discrete Math & Graph Theory (Prof. Glenn Tesler)

Lectured during discussion sections, held office hours, and graded assignments and exams.

Graduate Teaching Assistant Fall 2019–Spring 2020
University of Calgary, Calgary, AB
Undergraduate Physics Laboratory

Lectured during lab sections, guided experiments in optics and modern physics, and graded laboratory reports.

Student Fellow Fall 2018
Montana State University, Bozeman, MT
Honors Text & Critics

Assisted with class discussions and essay grading in a first-year honors seminar.